

MD. RAKIB HOSSAIN

Dhaka, Bangladesh | +880 1632 941507 | 000.mdrakibhossain@gmail.com
github.com/rakib4123 | linkedin.com/in/md-rakib-hossain-519818263

PROFESSIONAL SUMMARY

Computer Science and Engineering student specialising in machine learning and data science. Builds research-grade ML systems end to end: calibrated and uncertainty-aware models, computer-vision detection pipelines, custom datasets built from scratch, and honest evaluation that reports where a model fails as clearly as where it works. Comfortable shipping the result as a working application, and experienced in leading a technical team from concept to delivery.

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript/TypeScript.

Machine Learning: PyTorch, TensorFlow, scikit-learn, YOLO, OpenCV, SHAP

Data: pandas, NumPy, Matplotlib, Excel

Software and Tools: Git and GitHub, Jupyter, Google Colab, Claude Code, Antigravity, Roboflow, Next.js, NestJS, PostgreSQL

PROJECTS

RideGuard - Calibrated and Auditable ML Risk Framework | *Python, CatBoost, scikit-learn, SHAP*

- Crash-risk model combining a calibrated CatBoost scorer with a DBSCAN and KDE spatial prior: 0.973 macro-F1, calibration error cut 5.6x (ECE 0.043 to 0.008), conformal coverage 0.895 against a 0.90 target.
- Outperformed MLP, TabNet and Keras baselines; added ADWIN drift detection and SHAP explanations.
- Self-audit showed the model was fitting a near-deterministic label rather than real pre-crash signal, reported as the central finding. Deployed as a live web app with risk map and route-level alerts.

Research: github.com/rakib4123/RideGuard_V1 | *App:* github.com/rakib4123/Ride_Guard_WebApp | *Live:* ride-guard-web-app-web.vercel.app

DhakaNight - Low-Light Dataset Creation and Detection Benchmarking | *Python, PyTorch, YOLO, OpenCV, Roboflow*

- Built and annotated a 2,300-image low-light street dataset in Roboflow to cover a gap in existing detection data.
- Benchmarked CLAHE, Gamma, Zero-DCE and RetinexFormer against a YOLOv8 baseline (0.587 mAP@50): no enhancement method improved detection, RetinexFormer cost 5.4 points, and simply scaling the detector beat every preprocessing pipeline.

Code: github.com/rakib4123/Dhaka_Night

Drikon - Full-Stack E-Commerce Platform | *Next.js, NestJS, PostgreSQL, Prisma, Redis*

- Monorepo storefront and REST API with two-factor authentication, rotating JWTs, argon2id hashing and role-based access control, plus an admin panel for product and order management.

Live: drikon-web-vert1v.vercel.app | *Code:* github.com/rakib4123/drikon

PulseStone - Embedded Device and Controlled User Study | *ESP32-C3, Arduino C++, Python*

- Built a handheld breathing-guide device on an Arduino C++ finite-state machine with synchronized LED, haptic and display feedback.
- 15-participant study over 150 trials: 14.5x faster initiation, 7.49 s to 0.52 s ($t(14) = 70.5$, $p < 0.001$, Cohen's $d = 18.2$).

Code: github.com/rakib4123/PulseStone

LEADERSHIP AND ACTIVITIES

Founder and Team Lead, "Khepa Chakka" Robotics Team

2023 - Present

- Founded and lead a competitive robotics team, designing, building and programming four classes of robot (line-following, soccer, race, and battle), covering sensor integration, motor control, and real-time embedded logic.
- Coordinate members across mechanical design, electronics, and software against competition deadlines; led the team to six national and international podium finishes, including selection for Techfest, IIT Bombay.

AWARDS AND HONORS

- Winner, Cozmo Clench, Techfest IIT Bombay (Bangladesh Zonal), 2024 - one of Asia's largest science and technology festivals.
- Champion, Robot Soccer, AIUB CS Fest 2024; 1st Runner-up, Robot Soccer, AIUB Robotic Crew (ARC).
- 1st Runner-up (Nano Battle Bot) and 2nd Runner-up (Robo Race), National Robotics Championship 2025; 1st Runner-up, Robo Race, Technoxian Bangladesh National Round 2024.

EDUCATION

B.Sc. in Computer Science and Engineering, American International University-Bangladesh (AIUB)

Expected 2026

CGPA: 3.28 / 4.00

HSC, Dhaka Imperial College, 2021 | SSC, Dhaka Collegiate School, 2019 | **Languages:** Bangla (Native), English